



«Approved»
By the Dean
Syzdykova Z.A.
 _____ of _____ **2023**

Syllabus
Academic Year 2024 -2025

1. General information	
Course code	MC
Course title	Case study on data analytics
Degree cycle (level)/major	Graduate/Master in Computational Science
Year, semester	Year 1, Term 1
Number of credits	5
Language of delivery:	English
Prerequisites	Linear Algebra, Understanding of basic programming concepts and experience in at least one programming language, such as Python or MATLAB
Postrequisites	Numerical methods
Lecturer(s)	Ardashir Mohammadzadeh; a.mzadeh@astu.edu.cn Astana IT University, Expo, C1 block, 2nd floor, office C1.1.336
2. Goals, objectives and learning outcomes of the course	
1. Course description	This course provides a thorough exploration of data analytics concepts, methodologies, and tools, equipping students with the skills necessary to transform raw data into actionable insights.
2. Course goal(s)	- Introduction to Data Analytics: Understand the foundational principles of data analytics, including types of data. - Data Collection and Preparation: Learn techniques for gathering data from various sources and the critical steps involved in cleaning and preparing data for analysis. - Statistical Analysis: Gain proficiency in applying statistical methods to interpret data, including hypothesis testing, regression analysis, and correlation. - Predictive Analytics: Explore advanced analytics techniques such as machine learning algorithms to forecast future trends and behaviors based on historical data. - Data Visualization: Master the art of presenting data through effective visualization techniques using tools Matlab, or Python libraries. - Real-World Applications: Engage in hands-on projects and case studies that demonstrate how data analytics is applied across various industries, including finance, healthcare, marketing, and more.

3. Course objectives:	1. Develop Analytical Skills: Equip students with the ability to collect, clean, and prepare data for analysis, enabling them to identify patterns, trends, and insights that inform decision-making. 2. Apply Statistical Techniques: Enable students to use statistical methods and tools to analyze data sets effectively, interpret results, and draw meaningful conclusions that support business objectives. 3. Master Data Visualization and Communication: Teach students how to create compelling visual representations of data and communicate insights clearly to stakeholders, ensuring that findings are actionable and easily understood.
4. Skills & competences	1. Data Collection and Preparation: - Ability to gather data from various sources. - Proficiency in data cleaning and preprocessing techniques. 2. Statistical Analysis: - Understanding of key statistical concepts and methods (e.g., hypothesis testing, regression analysis). - Competence in using statistical software (e.g., R, Python, SPSS). 3. Data Visualization: - Skills in creating informative and visually appealing charts and graphs using tools like Matlab, or Matplotlib. - Ability to choose appropriate visualization techniques for different types of data. 4. Critical Thinking: - Capacity to analyze complex problems, evaluate evidence, and make data-driven decisions. - Ability to synthesize information from various sources to draw meaningful conclusions. 5. Communication Skills: - Proficiency in presenting data insights clearly and persuasively to diverse audiences. - Ability to write reports that effectively summarize findings and recommendations. 6. Project Management: - Skills in planning and executing data analysis projects, managing timelines, and collaborating with team members. - Ability to prioritize tasks and meet deadlines in a fast-paced environment.
5. Course learning outcomes:	1. Data Proficiency: - Students will be able to collect, clean, and prepare data from various sources for analysis, demonstrating proficiency in data preprocessing techniques.

	2. Statistical Analysis: - Students will apply appropriate statistical methods to analyze datasets, interpret results, and draw meaningful conclusions based on evidence. 3. Data Visualization: - Students will create effective visualizations using data visualization tools to communicate insights clearly and persuasively to diverse audiences. 4. Critical Thinking and Problem-Solving: - Students will demonstrate critical thinking skills by evaluating complex problems, synthesizing information from multiple sources, and making data-driven decisions.
6. Methods of assessment	Methods of assessment students' progress will be based on assignments (5 assignment per course). MidTerm, EndTerm and a Final examination. The further instruction will be provided in the course page in Moodle.
7. Reading list	Matlab-II: Computing, Programming and Data Analysis By: Yla Tausczik DATA ANALYSIS FROM SCRATCH WITH PYTHON By: Peters Morgan
8. Additional resources	Any other resources available on the Internet can be used by students to achieve the learning outcomes of the course. Materials of MIT University: https://ocw.mit.edu/help/faq-technology/
9. Course policy	Course and university policies include: Attendance is mandatory. Missing 30% of lessons will result in F (Fail) grade (or summer school). Late submissions are not accepted. No cheating, duplication, falsification of data, plagiarism, and crib Class work: The duration of each lecture and practical lesson is 50 minutes. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student would be dispensed from the classes. Being late on class: By the policy of this course, students who come late to class for more than 5 minutes are not allowed to get in to class and consequently, they will be marked as "absent" for the specific hour. <i>In the event of some extraordinary event, students should notify the teacher and request an extension of the deadline. If approved, a new date will be given to the student depending upon the circumstances.</i> Laptops and mobile devices can only be used for classroom purposes when directed by the teacher. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the teacher. Cheating and plagiarism are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data;

	4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using the learning management system (moodle.astanait.edu.kz). Contacting the Lecturer: students are welcome to arrange one-to-one meeting with the teacher during office hours to discuss the class.
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3. Course Content

#	Abbreviation	Meaning
1	TSIS	Teacher-supervised independent work
2	SIS	Students' independent work
4	PA	Practical assignment

3.1 Lecture, practical/seminar/laboratory session plans

Week No	Course Topic	Lectures (H/W)	Practice sessions	TSIS (H/W)	SIS (H/W)
1	Matlab Computing & Data Analysis; Constants and functions	2	3	1	9
2	Matlab Computing & Data Analysis; Operational manipulations	2	3	1	9
3	Matlab Computing & Data Analysis; Descriptive Statistics □	2	3	1	9
4	Matlab Computing & Data Analysis; Mathematical & Statistical Techniques	2	3	1	9
5	Data analysis in Python; Getting & Processing Data	2	3	1	9
6	Data analysis in Python; Data visualization	2	3	1	9

7	Data analysis in Python; Regression; supervised and supervised learning	2	3	1	9
8	Data analysis in Python; Classification; clustering	2	3	1	9
9	Data analysis in Python; Reinforcement learning	2	3	1	9
10	Data analysis in Python; Neural networks	2	3	1	9
Total hours: 150		20	30	10	90

3.2 List of assignments for Student Independent Study

Week No	Topics for independent study	Hours	Recommended sources	Form of submission
1-2	Data analysis in Python; Getting & Processing Data	6	[1], Chapter 7	Exercises
3-4	Data analysis in Python; Data visualization	6	[1], Chapter 8	Exercises
5-6	Data analysis in Python; Regression; supervised and supervised learning	6	[1], Chapter 9,10	Exercises
7-8	Data analysis in Python; Classification; clustering	6	[1], Chapter 11,12	Exercises
9-10	Data analysis in Python; Reinforcement learning	6	[1], Chapter 14	Exercises
1-10	Practical assignment	60	Books provided in the reading list, and other resources available on the internet	Assignment in Moodle
Total hours:		90		

4. Student performance evaluation system for the course

Period	Assignments	Number of points	Total
1 st attestation	Assignment 1 (by the end of Week 2)	15	100
	Assignment 2 (by the end of Week 3)	15	
	Assignment 3 (by the end of Week 4)	15	
	Assignment 4 (by the end of Week 5)	15	
	Mid Term (Week 5)	40	
2 nd attestation	Assignment 5 (by the end of Week 6)	15	100
	Assignment 6 (by the end of Week 7)	15	

	Assignment 7 (by the end of Week 8)	15	
	Assignment 8 (by the end of Week 9)	15	
	End Term (Week 10)	40	
Final exam	Final Assignment	100	100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final		100

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	
C-	1,67	60-64	Satisfactory
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	
F	0	0-24	Fail

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

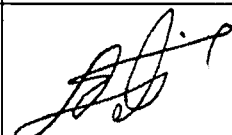
Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught

25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

3. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **TSIS (Teacher Supervised Student Independent Study)** - comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final exam:** The final exam for the course will be held in the form of a practical exam. More details will be provided at the time of final exam weeks. At the completion of the exam, all works must be submitted in the Learning Management System (moodle.astanait.edu.kz). No late submissions are allowed in the exam.

- Full name	Job title	Date	Sign
Ardashir Mohammadzadeh	Professor	02/09/2024	

Director

 Sergaziyev M.Zh.